

Static Cosmology

Data Exercise 12

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Topic

Optical Deflection and Medium Maps

Submission

Submit one ZIP archive containing report.pdf and the requested data table. Record the catalog and calibration versions in the report. The maximum file size is 25 MB.

Problems

1. Solve the first displayed relation used in Optical Deflection and Medium Maps. State all assumptions and conventions.

$$\alpha_{\perp} = \gamma \int \nabla_{\perp} \Phi ds / c^2$$

2. Estimate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$g_{obs} = g_{medium} + e_{intrinsic} + e_{PSF}$$

3. Analyze the third relation in two limiting regimes and explain the physical meaning of the result.

$$\Sigma_* = \operatorname{argmin}_{\Sigma} (\|K\Sigma - g\|_C^2 + \lambda_R \|L\Sigma\|^2)$$

Show enough intermediate work that another student can reproduce the calculation.